Mrp Steering

Component Design Document

1 Description

MRP steering attitude control law with integral feedback and reaction wheel momentum compensation. Steers the body rate toward a saturated rate command derived from the attitude error, and computes the commanded body torque from the rate tracking error, the reference motion, and the reaction wheel speeds. Wraps the MrpSteeringAlgorithm C++ algorithm via its C shim.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 5
- **Number of Invokee Connectors** - 3
- **Number of Invoker Connectors** - 2
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - *None*
- **Number of Parameters** - 12
- **Number of Events** - *None*
- **Number of Faults** - *None*
- **Number of Data Products** - 1
- **Number of Data Dependencies** - 2
- **Number of Packets** - *None*

3.2 Diagram



Figure 1: Mrp Steering component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Mrp Steering Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Reset_Tick_T_ Recv_Sync	recv_sync	Tick.T	-	1
Parameter_ Update_T_Modify	modify	Parameter_ Update.T	-	1

Connector Descriptions:

- **Tick_T_Recv_Sync** - Run the algorithm up to the current time.
- **Reset_Tick_T_Recv_Sync** - Zero the integral of the rate tracking error. Called on GNC state change so the integral does not carry over from the previous state.
- **Parameter_Update_T_Modify** - The parameter update connector.

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Mrp Steering Invoker Connectors

Name	Kind	Type	Return_Type	Count
Data_Product_ Fetch_T_Request	request	Data_Product_ Fetch.T	Data_Product_ Return.T	1
Data_Product_T_ Send	send	Data_Product.T	-	1

Connector Descriptions:

- **Data_Product_Fetch_T_Request** - Fetch a data product item from the database.
- **Data_Product_T_Send** - The data product invoker connector

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Mrp Steering Set Id Bases Parameters

Name	Type
`Data_Product_Id_Base`	`Data_Product_Types.Data_Product_Id_Base`
`Parameter_Id_Base`	`Parameter_Types.Parameter_Id_Base`

Parameter Descriptions:

- **`Data_Product_Id_Base`** - The value at which the component's data product identifiers begin.
- **`Parameter_Id_Base`** - The value at which the component's parameter identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains data dependencies that need their IDs mapped to valid data product IDs at initialization. Each data dependency also needs a stale limit provided in microseconds. If the timestamp of the data dependency is older than the reference time (usually the current time) minus the stale limit then the data dependency is considered too old to safely use. The `map_Data_Dependencies` procedure must be called with the following parameters:

Table 4: Mrp Steering Map Data Dependencies Parameters

Name	Type
`Attitude_Guidance_Id`	`Data_Product_Types.Data_Product_Id`
`Attitude_Guidance_Stale_Limit`	`Ada.Real_Time.Time_Span`
`Wheel_Speeds_Id`	`Data_Product_Types.Data_Product_Id`
`Wheel_Speeds_Stale_Limit`	`Ada.Real_Time.Time_Span`

Parameter Descriptions:

- **`Attitude_Guidance_Id`** - The external data product ID used to fetch the value for the 'Attitude_Guidance' data dependency.
- **`Attitude_Guidance_Stale_Limit`** - The stale limit used for the 'Attitude_Guidance' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.
- **`Wheel_Speeds_Id`** - The external data product ID used to fetch the value for the 'Wheel_Speeds' data dependency.

- **`Wheel_Speeds_Stale_Limit`** - The stale limit used for the 'Wheel_Speeds' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the MRP steering algorithm with the control period and the default parameter values. The `init` subprogram requires the following parameters:

Table 5: Mrp Steering Implementation Initialization Parameters

Name	Type	Default Value
`Control_Period`	`Basic_Types.Positive_Short_Float`	*None provided*

Parameter Descriptions:

- **`Control_Period`** - [s] Time between two algorithm updates. The rate error integral advances by this much per tick, so it must equal the rate at which the component is scheduled. Fixed by the assembly, so it is not a runtime parameter. The type keeps it greater than zero.

3.6 Commands

The Mrp Steering component has no commands.

3.7 Parameters

Parameters for the Mrp Steering component.

Table 6: Mrp Steering Parameters

Local ID	Parameter Name	Type	Default Value
0x0000 (0)	Proportional_Gain_K1	Packed_F32.T	(Value=>0.0)
0x0001 (1)	Cubic_Gain_K3	Packed_F32.T	(Value=>0.0)
0x0002 (2)	Omega_Max	Packed_F32.T	(Value=>0.02617994)
0x0003 (3)	Ignore_Outer_Loop_Feedforward	Packed_Boolean.T	(Value=>False)
0x0004 (4)	Derivative_Gain_Kd	Packed_F32.T	(Value=>0.0)
0x0005 (5)	Integral_Gain_Ki	Packed_F32.T	(Value=>0.0)
0x0006 (6)	Integral_Limit	Packed_F32.T	(Value=>0.0)
0x0007 (7)	Known_Torque_Pnt_B_B	Packed_F32x3.T	[0.0,0.0,0.0]
0x0008 (8)	Inertia	Packed_F32x9.T	[1.0,0.0,0.0,0.0,1.0,0.0,0.0,0.0,1.0]

0x0009 (9)	Rw_Spin_Axes	Packed_ F32x3_X4.T	[[1.0,0.0,0. 0],[0.0,1.0, 0.0],[0.0,0. 0,1.0],[0. 57735027,0. 57735027,0. 57735027]]
0x000a (10)	Rw_Inertias	Packed_ F32x4.T	[0.0,0.0,0. 0,0.0]
0x000b (11)	Wheel_Availability	Wheel_ Availability_ X4.T	[others= >Mrp_ Feedback_ Enums.Wheel_ Availability. Available]

Parameter Descriptions:

- **Proportional_Gain_K1** - [rad/s] Proportional gain applied to the MRP attitude error in the steering law. Must be finite and not negative.
- **Cubic_Gain_K3** - [rad/s] Cubic gain applied to the MRP attitude error in the steering saturation function. Must be finite and not negative.
- **Omega_Max** - [rad/s] Maximum rate command of the steering law. Must be finite and greater than zero.
- **Ignore_Outer_Loop_Feedforward** - [-] Whether the outer loop feedforward term, the rate of change of the steering rate command, is left out of the torque.
- **Derivative_Gain_Kd** - [Nms] Rate error feedback gain. Must be finite and not negative.
- **Integral_Gain_Ki** - [Nm] Integral feedback gain on the rate tracking error. Must be finite and not negative. Zero disables the integral term.
- **Integral_Limit** - [Nm] Anti-windup clamp on each component of the integrated rate tracking error. Must be finite and not negative.
- **Known_Torque_Pnt_B_B** - [Nm] Known external torque in body frame components. Must be finite.
- **Inertia** - [kg m^2] Spacecraft inertia about point B (3x3, row-major, 9 elements). Must be a valid inertia matrix.
- **Rw_Spin_Axes** - [-] Reaction wheel spin axes in body frame components, one unit vector per wheel slot. Each must be finite and within 1e-3 of unit length.
- **Rw_Inertias** - [kg m^2] Spin axis inertia of each reaction wheel. Each must be finite.
- **Wheel_Availability** - [-] Which reaction wheels contribute to the wheel momentum term. An unavailable wheel is left out. Every slot is configured, so a slot without a wheel is marked unavailable.

3.8 Events

The Mrp Steering component has no events.

3.9 Data Products

Data products for the Mrp Steering component.

Table 7: Mrp Steering Data Products

Local ID	Data Product Name	Type
0x0000 (0)	Control_Torque	Cmd_Torque_Body.T

Data Product Descriptions:

- **Control_Torque** - [Nm] Commanded control torque in body frame components.

3.10 Data Dependencies

Data dependencies for the Mrp Steering component.

Table 8: Mrp Steering Data Dependencies

Data Dependency Name	Type
Attitude_Guidance	Att_Guid.T
Wheel_Speeds	Rwa_Speeds.T

Data Dependency Descriptions:

- **Attitude_Guidance** - Attitude and rate tracking errors, the reference rate, and the reference angular acceleration, all in body frame components.
- **Wheel_Speeds** - [rad/s] Current speed of each reaction wheel.

3.11 Packets

The Mrp Steering component has no packets.

3.12 Faults

The Mrp Steering component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Mrp_Steering_Tests* Test Suite

This is a unit test suite for the Mrp Steering component

Test Descriptions:

- **Test** - Run the algorithm through the integral accumulation and reset sequence of the Python reference test to ensure the Ada to C to C++ integration is sound.
- **Test_Steering_Law_Variants** - Check the feedforward toggle, the saturation gains, and the maximum rate options against the Python reference model.
- **Test_Invalid_Parameter** - Ensure a staged configuration the algorithm would reject is refused at validation.
- **Test_Invalid_Data_Dependency** - Ensure a data dependency with the wrong identifier is treated as a wiring defect and fails the tick's assertion.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Att_Guid.T:

Table 9: Att_Guid Packed Record : 384 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Sigma_Br	Packed_F32x3.T	-	96	0	95
Omega_Br_B	Packed_F32x3.T	-	96	96	191
Omega_Rn_B	Packed_F32x3.T	-	96	192	287
Domega_Rn_B	Packed_F32x3.T	-	96	288	383

Field Descriptions:

- **Sigma_Br** - *No description provided.*
- **Omega_Br_B** - *No description provided.*
- **Omega_Rn_B** - *No description provided.*
- **Domega_Rn_B** - *No description provided.*

Cmd_Torque_Body.T:

Commanded control torque in body frame components, as produced by the MRP control laws.

Table 10: Cmd_Torque_Body Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Torque_Request_Body	Packed_F32x3.T	-	96	0	95

Field Descriptions:

- **Torque_Request_Body** - [Nm] Control torque requested

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 11: Data_Product Packed Record : 1096 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_Header.T	-	72	0	71	-

Buffer	Data_Product_ Types.Data_ Product_ Buffer_Type	-	1024	72	1095	Header. Buffer_Length

Field Descriptions:

- **Header** - The data product header
- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 12: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 13: Data_Product_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	48	63
Buffer_Length	Data_Product_ Types.Data_Product_ Buffer_Length_Type	0 to 128	8	64	71

Field Descriptions:

- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 14: Data_Product_Return Packed Record : 1104 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length

The_ Status	Data_ Product_ Enums. Fetch_ Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	-
The_ Data_ Product	Data_ Product. T	-	1096	8	1103	-

Field Descriptions:

- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Packed_Boolean.T:

Packed record holding a boolean type.

Table 15: Packed_Boolean Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Boolean	0 => False 1 => True	8	0	7

Field Descriptions:

- **Value** - The boolean value.

Packed_F32.T:

Single component record for holding packed 32-bit floating point number.

Table 16: Packed_F32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31

Field Descriptions:

- **Value** - The 32-bit floating point number.

Packed_F32x3.T:

Packed 3 element vector of 32-bit floats.

Table 17: Packed_F32x3 Packed Array : 96 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	3	96

Packed_F32x3_X4.T:

Array of 4 Packed_F32x3 elements, one three-vector per reaction wheel.

Table 18: Packed_F32x3_X4 Packed Array : 384 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Packed_F32x3.T	-	96	4	384

Packed_F32x4.T:

Table 19: Packed_F32x4 Packed Array : 128 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	4	128

Packed_F32x9.T:

Table 20: Packed_F32x9 Packed Array : 288 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	9	288

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 21: Parameter Packed Record : 792 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_ Header.T	-	24	0	23	-
Buffer	Parameter_ Types. Parameter_ Buffer_Type	-	768	24	791	Header.Buffer_ Length

Field Descriptions:

- **Header** - The parameter header
- **Buffer** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 22: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15

Buffer_Length	Parameter_Types. Parameter_Buffer_ Length_Type	0 to 96	8	16	23

Field Descriptions:

- **Id** - The parameter identifier
- **Buffer_Length** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 23: Parameter_Update Packed Record : 824 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_ Types. Parameter_ Table_Id	0 to 65535	16	0	15	-
Operation	Parameter_ Enums. Parameter_ Operation_ Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate 4 => Discard	8	16	23	-
Status	Parameter_ Enums. Parameter_ Update_ Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	-
Param	Parameter. T	-	792	32	823	-

Field Descriptions:

- **Table_Id** - The ID for the table that contains this parameter
- **Operation** - The parameter operation to perform.
- **Status** - The parameter return status.
- **Param** - The parameter that has been updated or fetched.

Rwa_Speeds.T:

Reaction Wheel Assembly (RWA) wheel speeds in rad/s.

Table 24: Rwa_Speeds Packed Record : 128 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Rwa_1	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31

Rwa_2	Short_Float	-3.40282e+38 to 3.40282e+38	32	32	63
Rwa_3	Short_Float	-3.40282e+38 to 3.40282e+38	32	64	95
Rwa_4	Short_Float	-3.40282e+38 to 3.40282e+38	32	96	127

Field Descriptions:

- **Rwa_1** - Wheel speed of RWA 1 in rad/s.
- **Rwa_2** - Wheel speed of RWA 2 in rad/s.
- **Rwa_3** - Wheel speed of RWA 3 in rad/s.
- **Rwa_4** - Wheel speed of RWA 4 in rad/s.

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 25: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^16) sub-seconds.

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 26: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces. Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

Wheel_Availability_X4.T:

Availability of each of the four reaction wheels, one byte per slot as the C boundary carries it. Values outside the enumeration fail type validation during parameter staging.

Table 27: Wheel_Availability_X4 Packed Array : 32 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Mrp_Feedback_Enums.Wheel_Availability.E	0 => Available 1 => Unavailable	8	4	32

5.3 Enumerations

The following section outlines any enumerations used in the component.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 28: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Mrp_Feedback_Enums.Wheel_Availability.E:

Whether a reaction wheel contributes to the wheel momentum term. Values match the C DeviceAvailability_c enumeration.

Table 29: Wheel_Availability Literals:

Name	Value	Description
Available	0	The wheel is running and its momentum is included.
Unavailable	1	The wheel is excluded from the momentum term.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 30: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.
Discard	4	Discard any parameters staged by this table that have not been updated.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 31: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.
Length_Error	3	Parameter length was not correct.